

Training at scale, and why big models are overconfident

1 Pretraining: guess the next word

The labels come free from the text itself, so models can learn from trillions of words.

2 More data, better guesses

Loss falls smoothly as data, model size and compute grow, with diminishing returns. That predictability is what made giant models a safe bet.

3 Tuning makes an assistant

Instruction tuning teaches it to follow requests; preference tuning teaches it to please raters. The second can cost calibration.

4 Accuracy hides overconfidence

Train too long: accuracy barely moves, log loss goes from 0.18 to 0.61, ECE from 0.019 to 0.072.

5 Why: squeezing the last bit of loss

On training data, pushing P towards 0 and 1 always lowers the loss, so the model learns to be sure.

6 Stop early, then calibrate

Temperature ($T = 3.7$) helped; Platt fixed the calibration (ECE 0.007). Stopping early beat both on log loss.